D11. Integrated Proof of Concept

End-to-end benchmark-based assessment demonstration using supplied audit reports.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

Demonstration acceptance trail

Proof step	Evidence produced	Completion
Load two reports	Source hashes and 22 chunks	Completed [E4]
Ask benchmark questions	16 structured question-level records	Completed [E5]
Validate evidence	Grounding, CRAG and review fields	Completed [E5]
Aggregate score	2-REPORT 100%; 4-REPORT 87.5%	Completed [E5]
Prevent premature release	Production readiness gate BLOCKED	Completed [E10]

What the proof demonstrates

The POC demonstrates an auditable benchmark-assessment workflow from source report through retrieval to an aggregate result. The key contribution is not merely that a model can answer questions: every answer can be inspected together with the evidence, configuration and resulting score. [E4][E5]

It intentionally does not demonstrate a production system. The proof set is too small, visual evidence is outside scope, and the release gate therefore protects the integrity of the submission conclusion. [E8][E10]

Demonstration objective

Demonstrate that the system can take an extracted cybersecurity audit report and benchmark questions, retrieve report evidence, issue structured decisions, and aggregate those decisions into an audit score with a reproducible trail. [E4][E5]

The demonstration uses two supplied JSON reports and not a synthetic substitute. It intentionally excludes image/figure context from the text-only submission scope. [E4][E6][E8]

Demonstrated configuration

Stage	Proof configuration	Recorded result
Input	2-report.json and 4-report.json; eight questions each	16 report-question pairs [E4][E6]
Evidence index	22 structure-aware chunks; Qwen3 embedding; hybrid top-8 retrieval	Per-query evidence saved [E4][E5]
Assessment	Qwen3-8B local Ollama; v2 prompt; schema enforcement; temperature 0	Answers and latencies saved [E4][E5]
Aggregation	70% pass threshold	Both reports produced PASS scores [E5]

Outcome

The run completed with no recorded execution errors. It achieved 81.25% answer accuracy, 87.50% Recall@8, 100% grounding under the implemented check, and 13.66-second mean latency. [E4][E5]

The reranker and claim verifier were disabled in this run to avoid competing with Qwen3-8B for the 6 GB laptop GPU. This is a documented hardware configuration, not evidence that those controls are unnecessary. [E4]

Reproduction and verification

Saved live artifacts: results/production_real_reports/1787846335_de0ded86a7db/. Offline functional demonstration: from rag/src, run python test_harness_mock.py. Focused parser, routing/citation and release-gate tests passed on 28 August 2026. [E4][E7]

Reproduction of the live run also requires the stated local model weights and Ollama service; it should remain in the approved air-gapped environment. [E4]

Proof boundary

This POC demonstrates an auditable integration path. It does not establish a production release, comparative model superiority, visual understanding, or generalisation beyond the two supplied reports.

Observed report-level assessment

Report	Questions	Yes / No / Insufficient	Score	POC decision
2-REPORT	8	8 / 0 / 0	100.0%	PASS at 70% threshold
4-REPORT	8	7 / 1 / 0	87.5%	PASS at 70% threshold
Production release	2 reports	12 Yes / 4 No / 0 IE labels	Not applicable	BLOCKED evidence gate

POC acceptance conclusion

The complete POC has been demonstrated on two actual supplied report files, not synthetic generated reports. It produces an audit score from the eight benchmark questions for each report and retains the question-level evidence needed to inspect both PASS outcomes. [E4][E5]

The production gate is intentionally more conservative than the POC scorer. A sample report can pass its selected benchmark while the system is still blocked from operational use because the evidence corpus lacks breadth, independent review and missing-evidence labels. That distinction is a strength of the submission, not a caveat hidden at the end. [E10]

Demonstration sequence for review

A reviewer can reproduce the POC narrative in five steps: identify one supplied JSON report; select one of the eight benchmark questions; inspect the retrieved chunk IDs and source paths; inspect the schema response and its cited evidence; then verify how the harness adds the Yes outcomes into the report score. The saved result files preserve every one of these stages. [E4][E5]

For 2-REPORT the harness records eight positive answers and therefore a 100% score. For 4-REPORT it records seven positive answers and one negative answer, producing 87.5%. Both clear the explicit 70% POC threshold. The figures are descriptive outcomes of the current rubric, not a statement that the reports are objectively secure. [E5]

The final POC acceptance condition is thus satisfied: the requested workflow exists, runs locally and is auditable. The intentionally separate release decision remains BLOCKED until evidence quality, not merely implementation completeness, improves. [E10]

Repository evidence cited

- [E4] results/production_real_reports/1787846335_de0ded86a7db/run_config.json and run.log: two-report production-parser run configuration and execution log.

- [E5] results/production_real_reports/1787846335_de0ded86a7db/{retrieval_metrics,question_level_llm_metrics,sota_faithfulness_metrics,audit_level_scores}.json: scored metrics and audit decisions.
- [E6] sample_data/{2-report.json,4-report.json,real_reports_benchmark_questions.json}: source reports and eight source-derived questions per report.
- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.